A Comparative Study of Data Generation Strategies for Imbalanced Classification

Synthetic, Simulated, and Augmented Data

Research question

Does changing the minority-class training distribution improve out-of-sample ranking?

Hypothesis under examination

Data-generation and reweighting strategies may help, but benefits are not guaranteed across data geometry, shift, and classifier choice.

Compared strategies

- **Baseline:** original training data, no intervention
- **Class weighting:** modify loss contribution, not observations
- **SMOTE:** interpolate minority-class neighbors
- **CTGAN:** learned conditional tabular generator
- **Oracle simulation:** minority observations from known synthetic mechanism

Oracle sampling is an experimental reference, not a deployable real-data oracle.

Five evaluation scenarios

Scenario	Characterization
S1	Overlapping class distributions
S2a	Nonlinear concentric rings, lower noise
S2b	Nonlinear concentric rings, higher noise
S3a	Test shift in a nuisance variable
S3b	Test shift in an informative variable

Learners

Logistic regression (LR) and random forest (RF).

Experimental protocol

Per evaluation seed

- Fit: 750 observations
- Validation: 250 observations
- Test: 2,000 observations
- 30 independent evaluation seeds

Leakage control

- Augment fitting split only
- Keep validation and test untouched
- CTGAN epochs chosen on separate development seeds
- Freeze epoch count at 10

$$5 \text{ scenarios} \times 30 \text{ seeds} \times 5 \text{ methods} \times 2 \text{ models} \times 2 \text{ threshold policies} = 3{,}000$$

Primary endpoint and inference

- **Average Precision (AP)**: primary, threshold-independent metric
- Paired within seed, scenario and learner: $\Delta AP = AP_{\text{method}} - AP_{\text{baseline}}$
- 40 contrasts: 5 scenarios $\times$ 2 learners $\times$ 4 alternatives
- Two-sided paired t tests; **Holm** correction across all 40 tests
- 95% confidence intervals shown in the analysis are **unadjusted**

Main finding

25/40

Holm-significant AP differences

0

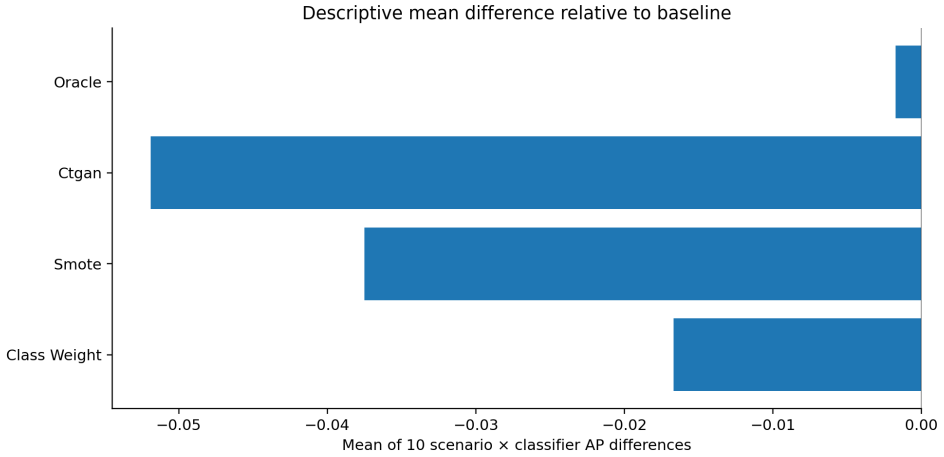
significant improvements over baseline

Direction of evidence

All 25 significant differences favored the **baseline**.

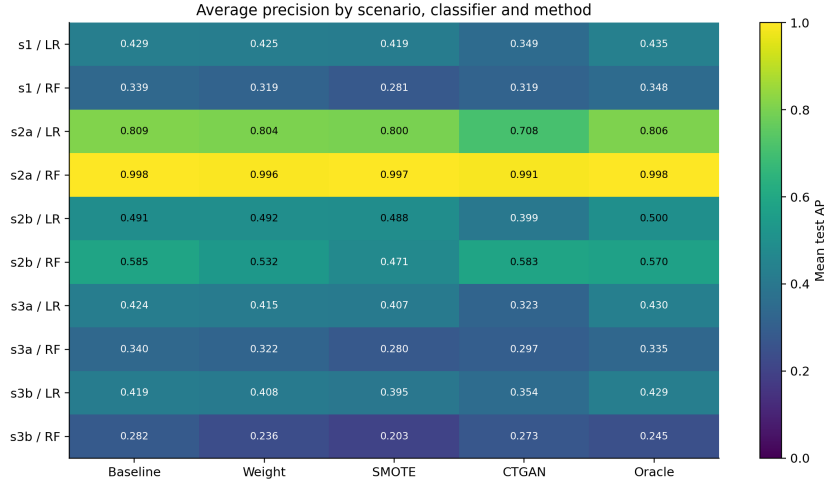
No conclusion of universal inferiority is warranted outside the evaluated conditions.

Average AP differences by strategy

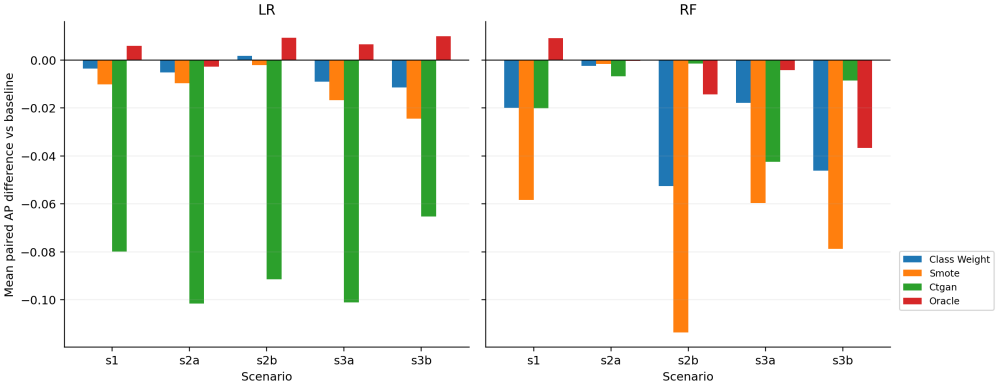


Descriptive means across ten scenario–classifier combinations; not pooled significance tests.

Where performance changes



Paired effects across conditions



Method-level summary

Strategy	Mean ΔAP	Significant losses / 10
Class weighting	−0.0167	6
SMOTE	−0.0375	9
CTGAN	−0.0519	8
Oracle simulation	−0.0018	2

Interpretation

Oracle simulation remained closest to baseline on average, while SMOTE and CTGAN frequently reduced AP. Responses were heterogeneous by scenario and classifier.

Why might augmentation hurt?

- Interpolation may create samples that do not respect complex class boundaries.
- A learned generator can reproduce training patterns without improving test ranking.
- Reweighting changes the optimization target, not necessarily ranking quality.
- Distribution shift can make additional minority observations less representative.

These are plausible explanations, *not* causally established mechanisms in this experiment.

Limitations and reproducibility

- Synthetic scenarios; no external real-world validation
- Fixed sample sizes, classifier choices and generation settings
- AP is primary; threshold-dependent outcomes are secondary
- Structural fidelity and runtime were **not** measured in the delivered result archive
- 30-seed paired analysis with 40-test multiplicity correction

Conclusions

Take-home message

More minority-class samples do not automatically mean better ranking performance.

- Evaluate augmentation against a strong untouched-data baseline.
- Separate model selection from confirmatory evaluation.
- Report negative findings and adjust for multiple comparisons.
- Validate on real-world datasets before generalizing.

Recent research: 2023–2026

- **Kotelnikov et al. (2023, ICML):** TabDDPM extends tabular generation beyond GANs to diffusion models.
- **D'souza et al. (2025, AAAI):** Explicitly modeling class-overlap regions can improve synthetic-data utility for imbalanced classification.
- **Won et al. (2026, Electronics):** Compared SMOTE, CTGAN, TVAE and Gaussian Copula on imbalanced real-world tabular data; reported gains in recall and F1.
- **Gündüz and Şahin (2026, Applied Sciences):** Compared SMOTE, CTGAN, TVAE and TabDDPM in a biological application.

These papers provide context, not direct replications: datasets, endpoints and protocols differ.

Positioning our findings against recent work

Why results need not agree

Reported gains in **recall/F1** do not imply gains in **average precision (AP)**; the metric, data geometry, learner and validation protocol all matter.

- Our study isolates five controlled scenarios with 30 paired seeds and Holm correction across 40 AP contrasts.
- Recent studies broaden the range of generators and real-world domains; they do not establish universal benefits.
- Future external validation should include real datasets, stronger generator tuning, runtime and fidelity measurements.

References: foundations and methods

- Chawla, N. V., et al. (2002). SMOTE: Synthetic Minority Over-sampling Technique. *JAIR*, 16, 321–357.
- He, H., and Garcia, E. A. (2009). Learning from Imbalanced Data. *IEEE TKDE*, 21(9), 1263–1284.
- Xu, L., et al. (2019). Modeling Tabular Data using Conditional GAN. *NeurIPS*.
- Saito, T., and Rehmsmeier, M. (2015). The Precision-Recall Plot Is More Informative than the ROC Plot. *PLOS ONE*, 10(3), e0118432.
- Holm, S. (1979). A Simple Sequentially Rejective Multiple Test Procedure. *Scandinavian Journal of Statistics*, 6(2), 65–70.
- Kotelnikov, A., et al. (2023). TabDDPM: Modelling Tabular Data with Diffusion Models. *ICML*, PMLR 202, 17564–17579.
- D'souza, A., M, S., and Sarawagi, S. (2025). Synthetic Tabular Data Generation for Imbalanced Classification: The Surprising Effectiveness of an Overlap Class. *AAAI*, 39(15), 16127–16134. DOI: 10.1609/aaai.v39i15.33771.
- Won, D.-H., Shin, K.-S., and Youm, S. (2026). Synthetic Data Augmentation for Imbalanced Tabular Data: A Comparative Study of Generation Methods. *Electronics*, 15(4), 883. DOI: 10.3390/electronics15040883.
- Gündüz, A. F., and Şahin, C. B. (2026). Synthetic Data Augmentation for Imbalanced Tabular Protein Subcellular Localization: A Comparative Study of SMOTE, CTGAN, TVAE, and TabDDPM Methods. *Applied Sciences*, 16(8), 3694. DOI: 10.3390/app16083694.
- Kurakova, A., and Homayouni, H. (2025). A comprehensive evaluation framework for synthetic medical tabular data generation. *Journal of Biomedical Informatics*, 171, 104939. DOI: 10.1016/j.jbi.2025.104939.

Thank you

Questions and discussion